

New Skills, Higher Pay? Evidence from Job Postings

David Koll Elmer Z. Li Marina M. Tavares

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The views expressed are those of the authors and do not necessarily represent the views of the IMF.

Motivation

- Labor markets are rapidly adopting **new skills**—AI, cloud computing, data science—yet we know little about how they are priced
- Key question: **Do employers pay more for newly emerging skills?**
- Understanding this matters for:
 - Wage inequality and skill-biased technological change
 - Worker training and education policy
 - Firm-level adoption of frontier technologies
- Challenge: separating the return to *new skills* from composition effects (firms, occupations, locations)

This Paper

Data

U.S. online job postings from **Lightcast**, 2010–2024 (~53M wage observations)

Measurement

New skill = rarely requested in 2010–11, prevalent thereafter
→ 2,046 new skills identified (of 33,000+); AI skills classified separately

Identification

Compare posted wages **within** the same occupation $\times$ industry $\times$ firm $\times$ county $\times$ year, controlling for total number of listed skills

Main finding: 2–4% wage premium for postings requiring new skills

Preview of Results

- ① **Baseline premium:** 2–4% for any new skill; rises steeply to $\sim 6\%$ for 4+ new skills
- ② **Heterogeneity:** IT skills $\sim 9\%$; AI developer $\sim 9\%$ vs. AI user $\sim 4\%$
- ③ **New job titles** alone carry *no* premium—only matter when paired with new skills
- ④ **Breakthrough patents:** linking new skills to upstream innovations yields substantially larger returns (IV: 11–49%)

Related Literature

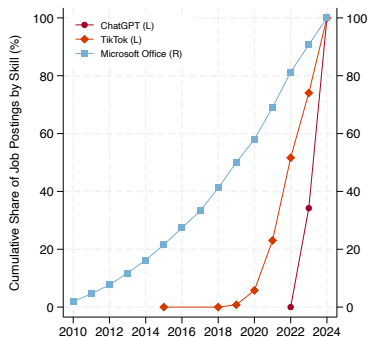
- **SBTC & task-based models:** Autor, Levy & Murnane (2003); Acemoglu & Restrepo (2019); Autor & Thompson (2025)
- **Online vacancies & new work:** Deming & Kahn (2018); Atalay et al. (2020); Hazell & Taska (2025)
- **AI and labor markets:** Acemoglu et al. (2022); Eloundou et al. (2024)
- **Patents & innovation:** Kelly et al. (2021); Autor (2024)

Our contribution

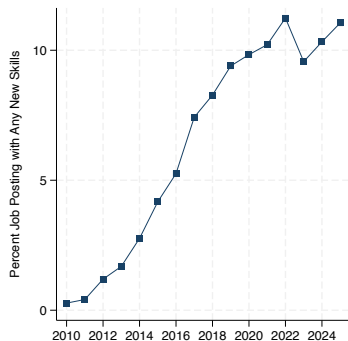
First to estimate **within-firm, within-cell wage premia** for newly emerging skills at scale—and link them to breakthrough patents via IV

Data: Lightcast & New Skill Definition

- Lightcast: ~220,000 online sources; 33,000+ unique skills tagged via NLP
- **New skill**: cumulative share of postings $< 1\%$ in 2010–11 base period, then rises above threshold
- 2,046 new skills meet criteria (appear in ≥ 100 postings)
- AI skills classified using Stanford AI Index + LLM-augmented review



(a) Cumulative Share of Job Postings by Skill (%)



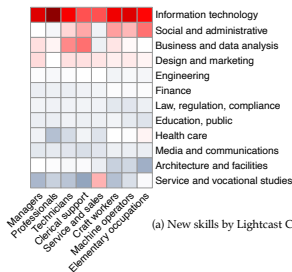
(b) Percent of Job Postings with Any New Skills

What Are New Skills? Examples by Occupation

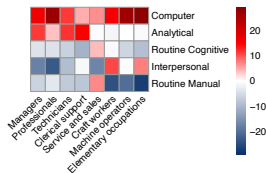
Managerial Occupations			Elementary Occupations		
Skill	1%	2%	Skill	1%	2%
Tableau (Business Intelligence)	2012	2013	Microsoft Edge	2018	2019
Google Workspace	2012	2013	POCO (C++ Library)	2012	2013
Power BI	2016	2017	Google Drive	2013	2014
Data Science	2013	2014	Class A/B UST Certification	2012	2012
Amazon Web Services	2012	2013	Sustainability Standards	2012	2013
Microsoft Azure	2012	2014	Go Server	2015	2018
Society for HR Management Cert.	2015	2015	Telecom. Device for the Deaf	2013	2014
Zoom (Video Conferencing Tool)	2012	2016	Mechatronics Certification	2013	2014

AI-related skills highlighted with shaded background.

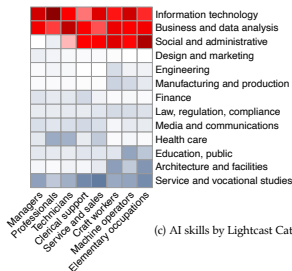
New Skill Intensity Across Categories



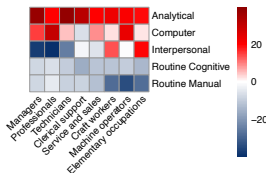
(a) New skills by Lightcast Category



(b) New skills by RNR



(c) AI skills by Lightcast Category



(d) AI skills by RNR

New skills cluster in **IT** and **Business/Data Analysis** domains.

Descriptive Statistics: Wage Regression Sample

	Full Wage Sample	Regression Sample
Number of postings	77,682,512	53,355,556
Mean hourly wage (in \$)	30.28	30.68
Average number of skills listed	12.2	12.5
Share with ≥ 1 new skill	8.5%	9.0%
Avg. new skills (conditional)	2.4	2.5
<i>By occupation type</i>		
High-skill	42.4%	43.6%
White-collar	23.0%	23.3%
Blue-collar	19.3%	19.6%
Low-skill	15.3%	13.6%
<i>By new skill category (share of all postings)</i>		
IT-related skills	3.9%	4.4%
Business and Data Analysis	2.1%	2.3%
Social / Administrative	1.6%	1.6%
All other skills	2.7%	2.7%

Empirical Strategy: Baseline Specification

$$\log(wage)_i = \beta \cdot NewSkill_i + \delta_{f \times t} + \delta_{occ \times t} + \delta_{reg \times t} + \delta_{stype \times t} + \delta_{\#skills} + \epsilon_i$$

- $NewSkill_i = 1$ if posting i lists ≥ 1 new skill
- **Firm $\times$ Year FE**: compare postings within the same firm-year
- **Occ $\times$ Year, Region $\times$ Year**: absorb broad occupation and local market trends
- **Pay-period $\times$ Year**: hourly vs. annually composition
- **Total # skills**: controls for posting complexity
- SE clustered at local labor market level (1,164 markets)

Saturated Specification

$$\log(wage)_i = \beta \cdot NewSkill_i + \delta_{f \times t} + \delta_{o \times ind \times c \times t} + \delta_{stye \times t} + \delta_{\#skills} + \epsilon_i$$

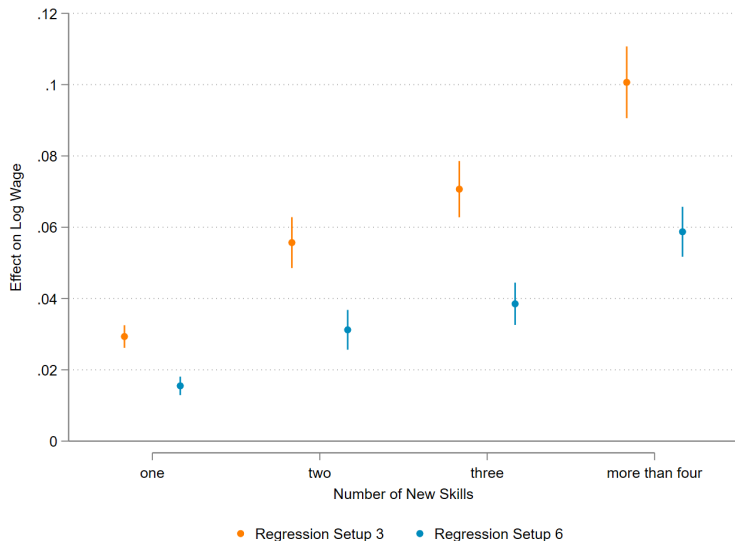
- Replace separate FEs with **Industry** $\times$ **Occupation** $\times$ **County** $\times$ **Year**
- Most demanding: “apples-to-apples” within the same detailed job-market cell
- Absorbs 7.5M fixed effects
- β captures: do firms pay more for new skills *within* the same firm-year and the same narrow labor market cell?

Main Results: Wage Premium to New Skills

	(1)	(2)	(3)	(4)	(5)	(6)
<i>Panel A: Presence of any New Skill</i>						
Any new skill	.0628*** (.0033)	.0421*** (.0022)	.0426*** (.0022)	.0340*** (.0023)	.0235*** (.0018)	.0234*** (.0018)
adj. R^2	0.650	0.739	0.754	0.791	0.829	0.837
<i>Panel B: Number of New Skills</i>						
One new skill	.0442*** (.0027)	.0288*** (.0016)	.0293*** (.0016)	.0234*** (.0018)	.0156*** (.0013)	.0155*** (.0013)
Two new skills	.0795*** (.0050)	.0552*** (.0036)	.0557*** (.0036)	.0437*** (.0033)	.0309*** (.0028)	.0312*** (.0028)
Three new skills	.1101*** (.0055)	.0690*** (.0041)	.0707*** (.0040)	.0600*** (.0036)	.0385*** (.0030)	.0385*** (.0030)
Four+ new skills	.1443*** (.0068)	.1034*** (.0052)	.1007*** (.0051)	.0815*** (.0041)	.0603*** (.0036)	.0587*** (.0036)
Firm $\times$ Year			X			X
Occ $\times$ Year / Reg $\times$ Year	X	X	X			
Ind $\times$ Occ $\times$ Reg $\times$ Year				X	X	X
Observations			53,355,556			

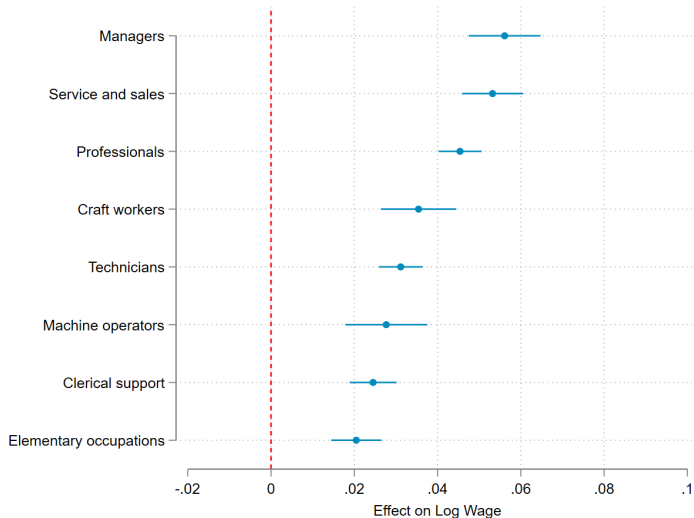
SE clustered at local labor market level. *** $p < 0.001$.

Wage Premium Rises Steeply with Number of New Skills



Baseline (col 3) vs. saturated (col 6) specification. Employers price the **intensity** of new-skill requirements, not merely their presence.

Heterogeneity by Occupation



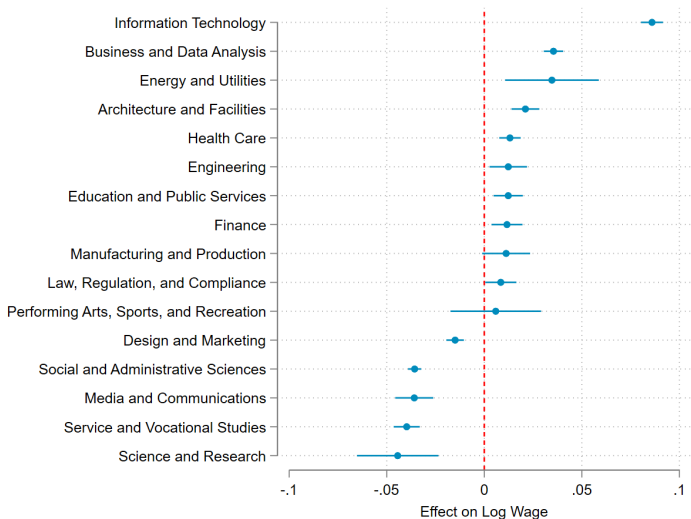
Managers ~6%, professionals ~4–5%, clerical/elementary ~2–3%. Premium positive across all occupation groups.

Heterogeneity by Income Terciles & Occupation Categories

	(1) Income Terciles	(2)	(3) Occupation Categories	(4)
Lowest Tercile / Blue collar	.0343*** (.0031)	.0230*** (.0021)	.0246*** (.0032)	.0224*** (.0031)
Middle Tercile / Low skill	.0249*** (.0022)	.0125*** (.0019)	.0311*** (.0039)	.0269*** (.0043)
Highest Tercile / High skill	.0564*** (.0031)	.0292*** (.0024)	.0434*** (.0027)	.0228*** (.0020)
— / White collar			.0453*** (.0028)	.0259*** (.0020)
Specification	Baseline	Saturated	Baseline	Saturated
Observations		53,355,556		

Highest-earning occupations benefit most; premium positive for all groups.

Heterogeneity by Skill Category



IT skills $\sim 9\%$, Business/Data Analysis $\sim 4\text{--}5\%$. Some categories *negative* (media, social/admin).

AI vs. Non-AI New Skills

	AI vs. Non-AI		AI Developer vs. User	
	(1) Baseline	(2) Saturated	(3) Baseline	(4) Saturated
Any new AI skill	.0737*** (.0035)	.0391*** (.0027)		
Any new AI User skill			.0396*** (.0037)	.0140*** (.0029)
Any new AI Developer skill			.0881*** (.0025)	.0572*** (.0019)
Any new non-AI skill	.0280*** (.0019)	.0158*** (.0015)	.0283*** (.0019)	.0160*** (.0015)
Observations	53,355,556			

- AI Developer skills: ~9% premium (e.g., PyTorch, TensorFlow)
- AI User skills: ~4% premium (e.g., ChatGPT, prompt engineering)
- Non-AI new skills: ~1.6%

Task Content of New Skills

Task Domain	Baseline	Saturated
Computer	.0543*** (.0025)	.0327*** (.0018)
Analytical	.0451*** (.0023)	.0205*** (.0018)
Routine Manual	.0417*** (.0032)	.0233*** (.0015)
Interpersonal	.0136*** (.0019)	.0050** (.0015)
Routine Cognitive	-.0049* (.0025)	.0173*** (.0030)

- **Computer** and **analytical** tasks most strongly rewarded
- Routine cognitive skills: close to zero—possible automation/standardization

New Job Titles vs. New Skills

	(1)	(2)	(3)	(4)	(5)	(6)
Emerging job title	−.0002 (.0011)	−.0076*** (.0009)	−.0071*** (.0009)	−.0033*** (.0009)	−.0038*** (.0007)	−.0032*** (.0007)
Any new skill	.0588*** (.0031)	.0389*** (.0020)	.0397*** (.0021)	.0323*** (.0022)	.0215*** (.0017)	.0215*** (.0017)
New skill × Emerging title	.0336*** (.0026)	.0324*** (.0021)	.0304*** (.0020)	.0167*** (.0019)	.0179*** (.0018)	.0166*** (.0017)
Firm × Year			X			X
Ind × Occ × Reg × Year				X	X	X
Observations			51,393,950			

- New titles alone: $\approx -0.3\%$ — **not a positive signal**
- New skills: $+2.2\%$; interaction: $+1.7\%$
- Wage premia driven by **skill content**, not rebranding

Breakthrough Patents: Motivation & Measurement

Goal

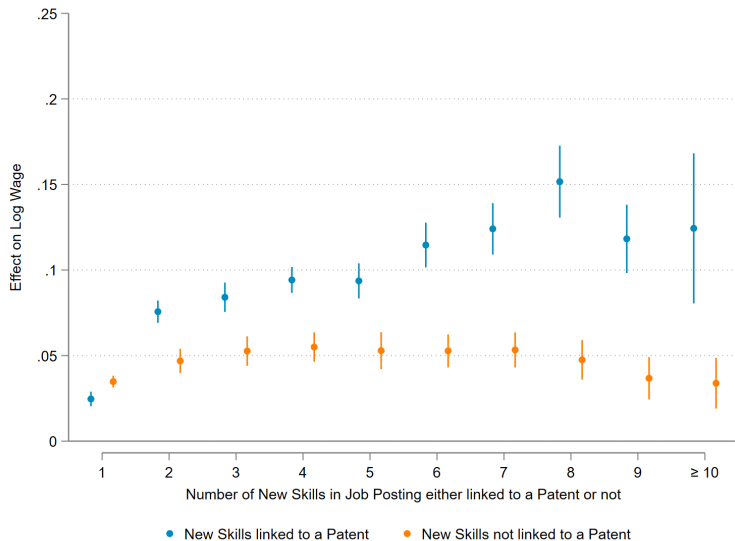
Link new skill demand to **upstream technological innovation**

- USPTO PatentsView data; “breakthrough” = top 10th percentile by Kelly et al. (2021) score
 - Low similarity to past patents (novelty) + high similarity to future patents (impact)
- Map patents → skills via OpenAI text-embedding-3-large; cosine similarity > 0.45

Key descriptive facts

- 97.9% of new-skill postings link to ≥ 1 patent
- But only **36.5%** have patent-linked *new* skills
- $\Rightarrow$ Patent-linked new skills offer a rare, informative source of variation

Wage Returns Rise with Patent-Linked New Skills



Non-linked new skills: ~4–6%. Patent-linked: rises to ~8% (2 skills) and >12% (7+ skills).

Monotonic gradient ties premia to frontier innovation.

Why Instrumental Variables?

- **Measurement error**: “new skill” classification is noisy → attenuation bias in OLS
- **Heterogeneity**: OLS averages over high-value (patent-linked) and low-value new skills
- **Solution**: instrument “any new skill” with count of breakthrough patents (2010–2016) linked to the posting’s skill bundle

Instrument

$$Z_i = \sum_{s \in \text{Skills}_i} \text{BreakthroughPatents}_{s, 2010-2016}$$

- Isolates variation driven by **upstream technological shocks**
- Pre-determined (patents from 2010–2016); wage sample is 2020–2024

IV Results: Wage Premium to New Skills

	(1) Baseline	(2) Saturated
<i>Panel A: IV — Unweighted Patent Count</i>		
Any new skill	.2198*** (.0249)	.1062** (.0353)
KP F-stat	1390.9	791.6
<i>Panel B: IV — Citation-Weighted Patent Count</i>		
Any new skill	.4811*** (.0763)	.4866*** (.1018)
KP F-stat	132.4	97.2
OLS benchmark (Table 1)	.0426	.0234

- IV $\gg$ OLS: consistent with **LATE** on high-value, frontier-driven skills
- Strong instruments (F-stats 97–1,391)
- OLS is a conservative **lower bound** for technology-driven returns

Conclusion

- ① New skills carry a **2–4% wage premium**, rising steeply with intensity (~6% for 4+ skills)
- ② Largest premia in **IT** (~9%) and **AI-developer** skills (~9% vs. 4% for AI user)
- ③ New job titles are **not priced** on their own—new **skills** are what matters
- ④ Returns anchored in **breakthrough technological innovation**; IV confirms and amplifies OLS

Implications

- Supporting acquisition of frontier technology skills can raise wages
- Skill content—not job labels—drives the market valuation of new work

Backup Slides

Robustness Summary

All exercises confirm the baseline results:

- **Alternative thresholds:** 0.5% and 2% cutoffs yield nearly identical premia
- **Time-based definitions:** skills appearing first after 2011 (unconditional or 3-year rolling window)—if anything, *larger* premia
- **Multi-way clustering:** SE clustered at $\text{LLM} \times (4\text{-digit ISCO} \times 6\text{-digit NAICS})$ —modestly larger SE, still highly significant
- **Excluding HR intermediaries:** premium attenuates slightly but remains positive and precise
- **Point wages only:** sample halves; premium attenuates to $\sim 1.4\%$ but monotonic gradient preserved

Descriptive Statistics (Full)

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Blue-collar	19.3%	19.6%
Low-skill	15.3%	13.6%
<i>By occupational income quintile</i>		
Quintile 1 (lowest)	16.3%	15.7%
Quintile 2	20.2%	20.2%
Quintile 3	29.7%	29.1%

Patent Descriptives

	All Postings	Postings with New Skills	
	Patents linked to any skill	Patents linked to any skill	Patents linked to new skills
Avg. # linked (new) skills	4.62	10.52	0.58
Share with ≥ 1 linked skill	88.4%	97.9%	36.5%
Avg. # associated patents	44.8	136.5	2.9
Share >10 patents	57.2%	87.7%	7.4%
Share >50 patents	24.1%	62.3%	0.8%
Share >100 patents	12.4%	42.9%	0.1%

Patent exposure is pervasive for overall skill bundles, but **rare for new skills**—providing quasi-experimental leverage for IV.